

YANTAO SHEN

Senior Applied Scientist @AWS Agentic AI, Bellevue, WA
ytshen@link.cuhk.edu.hk — (425) 623-0693 — <https://yantaoshen.github.io> — Google Scholar

RESEARCH INTERESTS

With over a decade of experience across Computer Vision and AI model training, my current research centers on LLM post-training—specifically SFT and RL for AI agents and the development of robust infrastructure to support large-scale model training and deployment.

EDUCATION

Multimedia Laboratory (MMLAB), The Chinese University of Hong Kong 2015 — 2020
Ph.D. in Electronic Engineering Supervisor: Prof. Xiaogang Wang, Prof. Hongsheng Li

Northeastern University (CN) 2011 — 2015
B.Eng. in Control Science and Engineering

WORK EXPERIENCE

AWS Fundamental Research/Agentic AI Bellevue, WA
Senior Applied Scientist April 2024 - Present

- **Reinforcement Learning for LLMs and Multimodal Models:** Led research on applying RL techniques to improve model capabilities across language and vision domains, including (1) During RL training, conducting distillation between teacher and student models based on Trust Region Ratio for efficient reasoning distillation, balancing exploration, exploitation, and imitation; and (2) Group Relative Policy Optimization (GRPO) based diffusion models for text-guided object spatial editing, reducing the need for costly paired datasets. The related publications are
 - RL-aware Knowledge Distillation for LLM Reasoning (submitted to ICML 2026);
 - Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes (submitted to CVPR 2026) (arxiv)
- **Multi-LLM model Inference and Reasoning Optimization:** Led and contributed to research on improving LLM inference performance through (1) token-level logit fusion for dynamic multi-LLM ensembling using sparse supervision, outperforming routing and merging baselines; and (2) design a lightweight neural probe on hidden states for parallel multi-criteria reasoning verification, achieving $3\text{-}5\times$ speedup over LLM/PRM-based approaches and still obtained on-par accuracies. The related publications are
 - DLLG: Dynamic Logit-Level Gating of LLM Experts (submitted to ICML 2026);
 - Accelerating Reasoning Through Step Speculation with Parallel Hidden States Verification (submitted to ICML 2026)
- **Training Infra Development on Trainium for States Space Model:** Served as a core contributor for state space model training development on Trainium, an AWS designed chip for AI training and inference; Successfully reduced training costs and accelerated SSM performance on Trainium through the strategic application of advanced parallelism techniques, including Tensor, Sequence, and ZeRO-Data Parallelism. The related publications are
 - B'MOJO: Hybrid State Space Realizations of Foundation Models with Eidetic and Fading Memory (NeurIPS 2024)
 - Open-source code: <https://github.com/awsllabs/state-space-models-neuron>

AWS, Rekognition Seattle, WA
Applied Scientist Aug 2021 – April 2024
Applied Scientist Intern Jun 2019 – Dec 2019

- **Model-Compatible Training.** Pioneered a backward-compatible training (BCT) framework for vision models, establishing a new research track in model compatibility. This innovation (1) allows legacy embeddings to remain compatible with upgraded models—eliminating the need for costly re-embedding. (2) Reducing negative flip errors in facial recognition classification. The related publications are
 - Towards Backward-Compatible Representation Learning (CVPR 2020, Oral),
 - Elodi: Ensemble Logit Difference Inhibition for Positive-Congruent Training (T-PAMI)
 - US patent: Backward compatible and backfill-free image search system
 - Open-source code: openBCT, regression-constraint-model-upgrade

- **Multi-CV-task joint training** Led research into Multi-Task Learning architectures for CV models, achieving performance parity or gains over task-specific models while consolidating multiple models into a single unified backbone. Developed and deployed a joint training framework for AWS Rekognition's Face Attribution Model, significantly reducing model engineering overhead and inference costs. The related publication are
 - Musketeer: Joint Training for Multi-task Vision Language Model with Task Explanation Prompts
 - AWS Rekognition Face doc

Tencent, ARC Lab
Applied Researcher

Shenzhen, Guangdong, China
 July 2020 – July 2021

- **Compatible Model Deployment for Tencent Short Video Platform.** Implemented backward-compatible algorithms for Tencent Short Video platform to enable seamless model upgrades. Reduced system downtime and re-computation costs for billions of video embeddings. The related publications are
 - Towards Universal Backward-Compatible Representation Learning (IJCAI 2022, Long Oral)
 - Hot-Refresh Model Upgrades with Regression-Alleviating Compatible Training (ICLR 2022)

Recent Publication

Most recent publications are introduced in the above section. Here are some publications between 2020-2023, For my early publications before 2020, please refer to my Google Scholar page.

ELODI: Ensemble Logit Difference Inhibition for Positive-Congruent Training T-PAMI
 Y. Zhao, **Y. Shen**, Y. Xiong, S. Yang, W. Xia, Z. Tu, B. Schiele, S. Soatto

Towards Universal Backward-Compatible Representation Learning IJCAI 2022 (Long Oral)
 B. Zhang, Y. Ge, **Y. Shen**, Y. Li, C. Yuan, X. Xu, Y. Wang, Y. Shan.

Hot-Refresh Model Upgrades with Regression-Alleviating Compatible Training in Image Retrieval ICLR 2022
 B. Zhang, Y. Ge, **Y. Shen**, S. Su, F. Wu, C. Yuan, X. Xu, Y. Wang, Y. Shan.

Towards Backward-Compatible Representation Learning CVPR 2020 (Oral)
Y. Shen, Y. Xiong, W. Xia, S. Soatto

SKILLS

- **Programming:** Python, L^AT_EX, Matlab
- **DL frameworks:** Caffe, PyTorch, MXNet.

Invited Talk and Media Press

- **Towards Backward Compatible Representation Learning** (link), *Tutorial on Cross-Model Compatibility, ICCV, 2021, Vitural*
- **VentureBeat Coverage:** *Amazon proposes technique for improving computer vision models without reprocessing* (link)